

DESIGN AND IMPLEMENTATION OF A SECURE ENTERPRISE NETWORK USING VLANs AND INTER-VLAN ROUTING

Prepared By

SAMUEL OCHOCHÉ PETER
(FE/25/80300835222)

Cybersecurity Student
3MTT Cybersecurity Program

Project Title

Configuration of IP Addressing and Virtual Local Area Networks (VLANs) in a Simulated Enterprise Network Environment

Project Focus

- VLAN Design and Implementation
- Network Segmentation
- IP Addressing Configuration
- Inter-VLAN Routing
- Access Control Lists (ACLs)
- Network Security and Traffic Isolation
- Enterprise Network Management

Tools Used

- Cisco Packet Tracer
- Cisco Router
- Cisco Switch
- Windows Command Prompt
- Networking and Security Configuration Tools

Date: 8th June, 2026

INTRODUCTION

Modern organizations rely heavily on computer networks to support communication, data sharing, and access to critical business resources. As enterprise networks continue to grow in size and complexity, network administrators must implement effective strategies to improve security, performance, scalability, and manageability. One of the most widely used techniques for achieving these objectives is the implementation of Virtual Local Area Networks (VLANs).

A Virtual Local Area Network (VLAN) is a logical segmentation of a physical network that allows devices to be grouped according to business functions rather than physical location. VLANs help reduce broadcast traffic, improve network performance, enhance security, and simplify network administration. By separating departments into different VLANs, organizations can control communication between users and protect sensitive resources from unauthorized access.

In addition to VLAN implementation, organizations often require communication between separate network segments. This is achieved through Inter-VLAN Routing, a technique that enables devices in different VLANs to communicate while maintaining network segmentation. Inter-VLAN Routing provides controlled connectivity between departments and allows administrators to enforce security policies using routing and access control mechanisms.

This project focuses on the design and implementation of a secure enterprise banking network using Cisco Packet Tracer. The network is divided into multiple departments, including Customer Service, Accounts, Information Technology (IT), and ATM/Server Infrastructure. Each department is assigned a dedicated VLAN and IP subnet to achieve logical separation and improve security.

To further strengthen the network, Access Control Lists (ACLs) were implemented to restrict unauthorized communication between departments. Security policies were designed to prevent Customer Service users from accessing IT resources while allowing authorized communication between the Accounts Department and the ATM Server network.

The project demonstrates the practical application of VLAN segmentation, trunk configuration, IP addressing, Inter-VLAN Routing, and ACL-based security controls within a simulated enterprise environment. Various verification and testing procedures were performed to ensure that network communication, segmentation, and security requirements were successfully achieved.

The knowledge and skills demonstrated in this project are directly applicable to modern enterprise networks, financial institutions, government agencies, and other organizations that require secure, scalable, and well-managed network infrastructures.

PROJECT GOALS

The primary objective of this project was to design and implement a secure enterprise network using Virtual Local Area Networks (VLANs) and Inter-VLAN Routing. The project was developed in Cisco Packet Tracer to simulate a real-world banking environment where different departments require network segmentation, controlled communication, and enhanced security.

Specific objectives of the project include:

- Design a functional enterprise network topology using Cisco Packet Tracer.
- Implement VLANs to logically separate departments and reduce unnecessary broadcast traffic.
- Configure VLAN names and assign switch ports according to departmental requirements.
- Configure IP addressing schemes for all end devices and network interfaces.
- Implement Inter-VLAN Routing using the Router-on-a-Stick technique.
- Configure trunk links to allow multiple VLAN traffic between the switch and router.
- Apply Access Control Lists (ACLs) to enforce security policies and restrict unauthorized communication.
- Verify connectivity through ping tests and network troubleshooting tools.
- Demonstrate how network segmentation improves security, performance, and manageability in an enterprise environment.
- Gain practical experience with enterprise networking concepts used in modern organizations.

ASSET, THREAT, AND VULNERABILITY ANALYSIS

Asset Identification

Assets are valuable resources that require protection within the network environment. The following assets were identified in this project:

Network Infrastructure Assets

- Cisco Router
- Cisco Switch
- Trunk Links
- VLAN Interfaces

End User Assets

- Customer Service Department Computers
- Accounts Department Computers
- IT Department Computers

Critical Business Assets

- ATM Server
- Financial Records
- Customer Information
- Internal Banking Systems

Network Services

- Inter-VLAN Routing Services
- IP Addressing Services
- Internal Communications

Threat Identification

The following threats were identified as potential risks to the banking network:

Unauthorized Access

Attackers may attempt to access sensitive departments such as IT or Server networks without authorization.

Internal Threats

Employees may intentionally or unintentionally access restricted systems or confidential information.

Malware Propagation

A malware infection in one department may spread across the network if proper segmentation is not implemented.

Network Reconnaissance

Attackers may scan the network to identify active hosts, services, and vulnerabilities.

Denial of Service (DoS)

Attackers may attempt to overwhelm network resources and disrupt operations.

VLAN Hopping Attacks

Attackers may attempt to bypass VLAN boundaries and gain access to unauthorized network segments.

Data Theft

Sensitive customer information and banking records may be targeted for theft.

Vulnerability Identification

Before VLAN implementation, the network could be vulnerable to:

- Flat Network Architecture
- Excessive Broadcast Traffic
- Lack of Traffic Isolation
- Weak Access Control Mechanisms
- Unrestricted Communication Between Departments
- Increased Attack Surface

Risk Mitigation Measures

To reduce identified risks, the following security controls were implemented:

- VLAN Segmentation
- Inter-VLAN Routing
- Access Control Lists (ACLs)
- Departmental Traffic Isolation
- Controlled Network Access
- Secure Network Architecture

The implementation of these controls significantly improves confidentiality, integrity, and availability of network resources.

IP ADDRESSING PLAN

To ensure proper communication and network organization, each department was assigned a unique IP subnet and VLAN. This approach simplifies management and improves security through network segmentation.

Table 1: VLAN and IP Addressing Scheme

Table 1: VLAN and IP Addressing Scheme

VLAN ID	Department	Network Address	Default Gateway
VLAN 10	Customer Service	192.168.10.0/24	192.168.10.1
VLAN 20	Accounts	192.168.20.0/24	192.168.20.1
VLAN 30	IT Department	192.168.30.0/24	192.168.30.1
VLAN 40	ATM Server	192.168.40.0/24	192.168.40.1

Table 2: Device IP Address Allocation

Table 2: Device IP Address Allocation

Device	VLAN	IP Address
Customer PC1	VLAN 10	192.168.10.2
Customer PC2	VLAN 10	192.168.10.3
Customer PC3	VLAN 10	192.168.10.4
Accounts PC1	VLAN 20	192.168.20.2
Accounts PC2	VLAN 20	192.168.20.3

Each device was configured with:

- IP Address
- Subnet Mask (255.255.255.0)
- Default Gateway

This configuration enabled communication within the same VLAN and through the router for inter-VLAN communication.

NETWORK TOPOLOGY DESIGN

The network topology for this project was designed to represent a small enterprise banking environment using Cisco Packet Tracer. The design consists of one router, one switch, multiple departmental computers, and a server/ATM network. The aim of the topology is to demonstrate how VLANs can be used to segment an enterprise network into separate departments while allowing controlled communication through Inter-VLAN Routing.

The topology uses a star-based network design where all end devices are connected to a central switch. The switch is connected to a router through a trunk link. The router performs Inter-

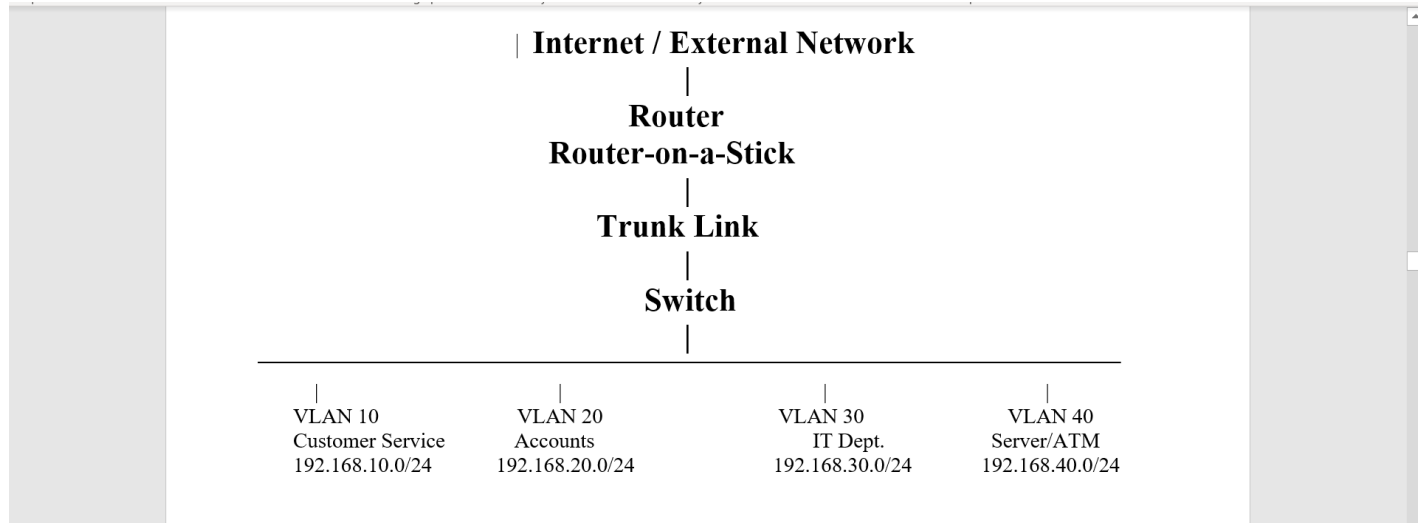
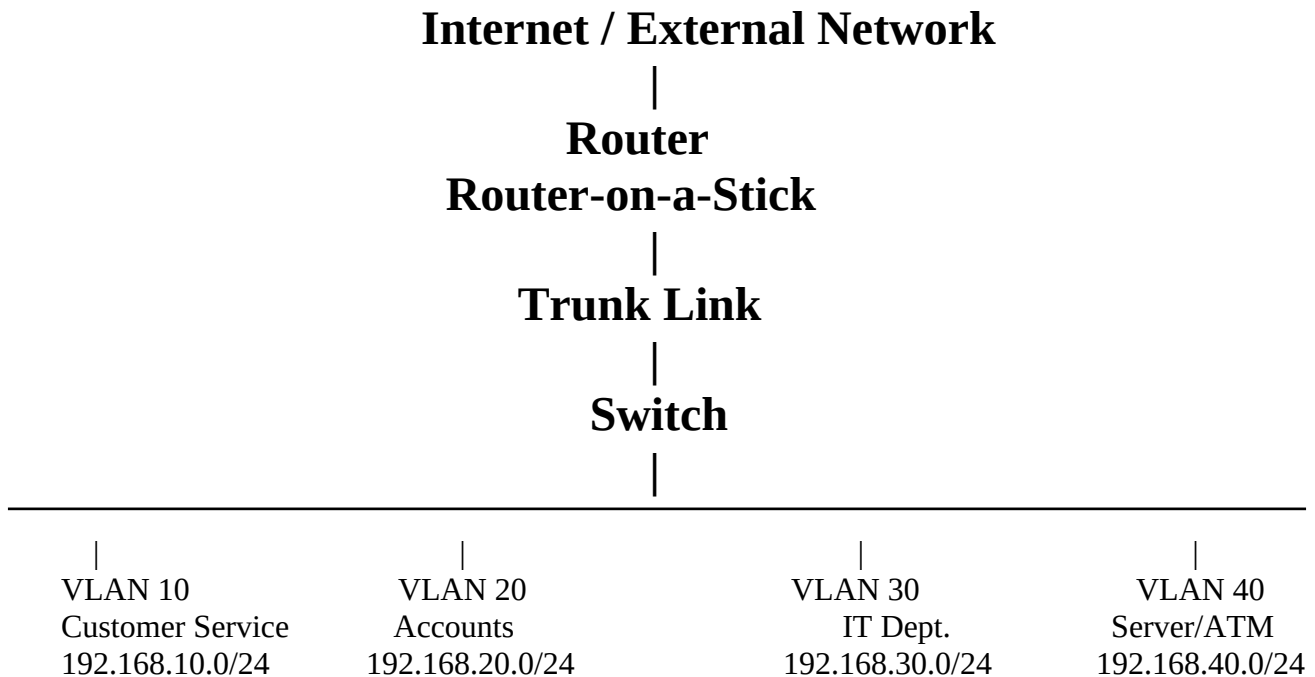
VLAN Routing using the Router-on-a-Stick method, which allows devices in different VLANs to communicate through router subinterfaces.

The network was divided into four major VLANs:

VLAN ID	VLAN Name	Department	Network Address
VLAN 10	Customer_Service	Customer Service Department	192.168.10.0/24
VLAN 20	Accounts	Accounts Department	192.168.20.0/24
VLAN 30	IT	IT Department	192.168.30.0/24
VLAN 40	Server	Server/ATM Network	192.168.40.0/24

VLAN ID	VLAN Name	Department	Network Address
VLAN 10	<u>Customer_Service</u>	Customer Service Department	192.168.10.0/24
VLAN 20	Accounts	Accounts Department	192.168.20.0/24
VLAN 30	IT	IT Department	192.168.30.0/24
VLAN 40	Server	Server/ATM Network	192.168.40.0/24

Logical Network Diagram:



The router is responsible for routing traffic between VLANs. Each VLAN has a unique default gateway configured on the router subinterface. The switch is responsible for VLAN creation, VLAN naming, port assignment, and trunk configuration.

Topology Components:

Device	Function
Router	Provides Inter-VLAN Routing
Switch	Provides VLAN segmentation and port assignment
Customer Service PCs	End devices in VLAN 10
Accounts PCs	End devices in VLAN 20
IT Department PCs	End devices in VLAN 30
ATM/Server	Critical resource in VLAN 40



Device	Function
Router	Provides Inter-VLAN Routing
Switch	Provides VLAN segmentation and port assignment
Customer Service PCs	End devices in VLAN 10
Accounts PCs	End devices in VLAN 20
IT Department PCs	End devices in VLAN 30
ATM/Server	Critical resource in VLAN 40



The topology was designed to improve network security, reduce broadcast traffic, and separate departments based on their roles. For example, Customer Service users were placed in VLAN 10, Accounts users in VLAN 20, IT users in VLAN 30, and critical server/ATM resources in VLAN 40.

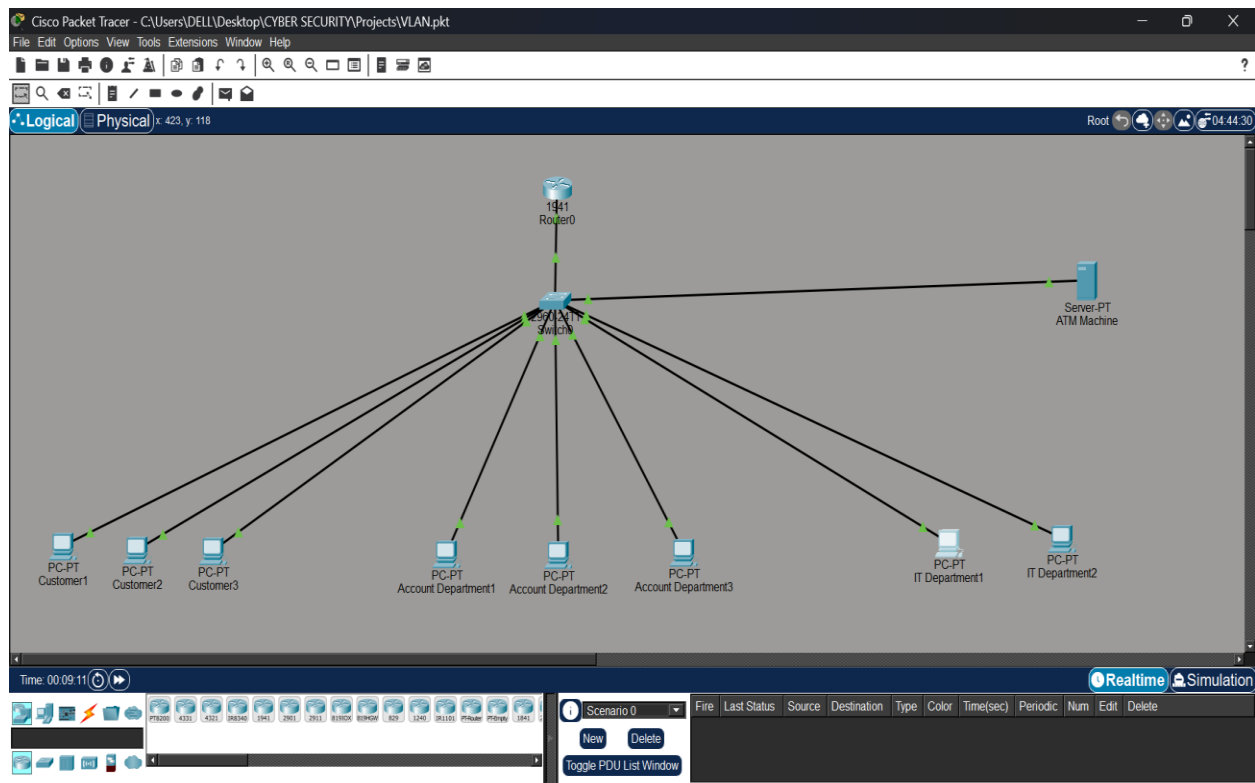
This design supports the principle of network segmentation because each department operates in a separate broadcast domain. Without Inter-VLAN Routing, devices in one VLAN cannot

communicate with devices in another VLAN. Controlled communication is enabled only through the router, where security rules such as Access Control Lists (ACLs) can be applied.

Security Benefits of the Topology:

- Separates departments into different VLANs.
- Reduces unnecessary broadcast traffic.
- Prevents unrestricted access between departments.
- Allows security policies to be applied at the router.
- Protects sensitive departments such as IT and Server/ATM networks.
- Improves network performance and manageability.

Figure 1: Network Topology



VLAN DESIGN AND CONFIGURATION

Virtual Local Area Networks (VLANs) were implemented to logically segment the enterprise network into separate departments. VLAN segmentation reduces broadcast traffic, improves network performance, and enhances security by isolating users and resources into distinct network segments.

Four VLANs were created to represent the various departments within the banking environment:

Table 3: VLAN Design

VLAN ID	VLAN Name	Department	Network Address
VLAN 10	CUSTOMER	Customer Service	192.168.10.0/24
VLAN 20	ACCOUNTS	Accounts Department	192.168.20.0/24
VLAN 30	IT	Information Technology	192.168.30.0/24
VLAN 40	SERVER	ATM/Server Network	192.168.40.0/24

The switch ports were assigned to their respective VLANs to ensure proper departmental separation.

Port Assignment:

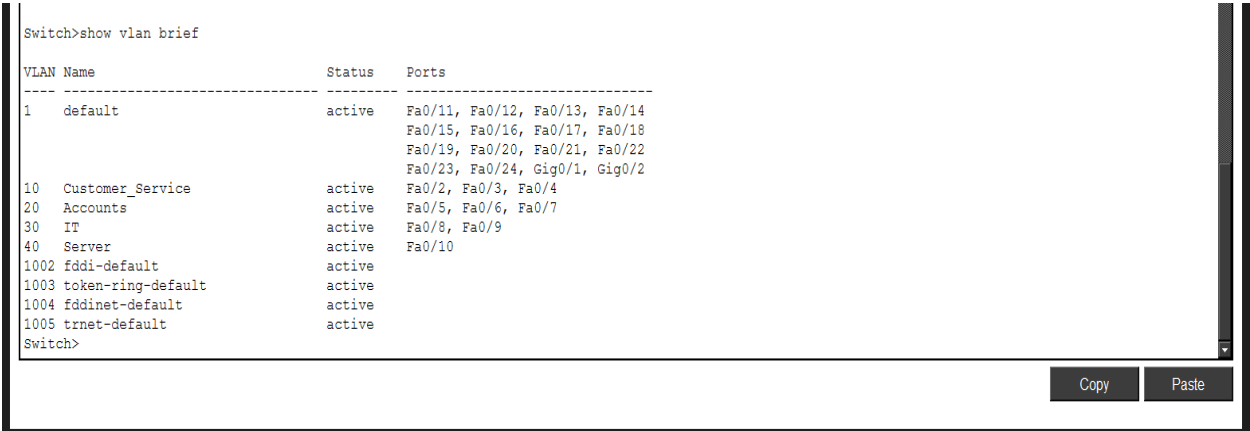
- Fa0/2 – Fa0/4 → VLAN 10 (Customer Service)
- Fa0/5 – Fa0/7 → VLAN 20 (Accounts)
- Fa0/8 – Fa0/9 → VLAN 30 (IT)
- Fa0/10 → VLAN 40 (ATM/Server)

The VLAN configuration was verified using the command:

```
show vlan brief
```

The output confirmed that all VLANs were successfully created and that the correct ports were assigned to their designated VLANs.

Figure 2: VLAN Configuration (show vlan brief)



IP ADDRESSING PLAN

An organized IP addressing scheme was developed to provide unique network segments for each VLAN. Each department was assigned a dedicated subnet, ensuring efficient traffic management and simplified network administration.

Table 4: VLAN IP Addressing Scheme

VLAN ID	Department	Network Address	Default Gateway
VLAN 10	Customer Service	192.168.10.0/24	192.168.10.1
VLAN 20	Accounts	192.168.20.0/24	192.168.20.1
VLAN 30	IT Department	192.168.30.0/24	192.168.30.1
VLAN 40	ATM/Server Network	192.168.40.0/24	192.168.40.1

VLAN ID	Department	Network Address	Default Gateway
VLAN 10	Customer Service	192.168.10.0/24	192.168.10.1
VLAN 20	Accounts	192.168.20.0/24	192.168.20.1
VLAN 30	IT Department	192.168.30.0/24	192.168.30.1
VLAN 40	ATM/Server Network	192.168.40.0/24	192.168.40.1

Device Address Allocation:

Device	VLAN	IP Address
Customer PC1	VLAN 10	192.168.10.2
Customer PC2	VLAN 10	192.168.10.3
Customer PC3	VLAN 10	192.168.10.4
Accounts PC1	VLAN 20	192.168.20.2
Accounts PC2	VLAN 20	192.168.20.3
Accounts PC3	VLAN 20	192.168.20.4
IT PC1	VLAN 30	192.168.30.2
IT PC2	VLAN 30	192.168.30.3
ATM Server	VLAN 40	192.168.40.2



Device	VLAN	IP Address
Customer PC1	VLAN 10	192.168.10.2
Customer PC2	VLAN 10	192.168.10.3
Customer PC3	VLAN 10	192.168.10.4
Accounts PC1	VLAN 20	192.168.20.2
Accounts PC2	VLAN 20	192.168.20.3
Accounts PC3	VLAN 20	192.168.20.4
IT PC1	VLAN 30	192.168.30.2
IT PC2	VLAN 30	192.168.30.3
ATM Server	VLAN 40	192.168.40.2

Each device was configured with:

- IP Address
- Subnet Mask (255.255.255.0)
- Default Gateway

This addressing plan ensured that devices could communicate within their VLAN and reach other authorized VLANs through the router.

INTER-VLAN ROUTING CONFIGURATION

Since VLANs create separate broadcast domains, devices located in different VLANs cannot communicate directly. To enable communication between departments while maintaining network segmentation, Inter-VLAN Routing was implemented using the Router-on-a-Stick technique.

Router-on-a-Stick is a method where a single physical router interface is divided into multiple logical subinterfaces. Each subinterface is assigned to a VLAN and acts as the default gateway for that VLAN.

The following router subinterfaces were configured:

Subinterface	VLAN	Gateway Address
G0/0.10	VLAN 10	192.168.10.1
G0/0.20	VLAN 20	192.168.20.1
G0/0.30	VLAN 30	192.168.30.1
G0/0.40	VLAN 40	192.168.40.1

IEEE 802.1Q encapsulation was used to identify VLAN traffic.

A trunk connection was configured between the switch and router to carry traffic from all VLANs across a single physical link.

The trunk configuration was verified using:

```
show interfaces trunk
```

The router configuration was verified using:

show ip interface brief

The successful output confirmed that all subinterfaces were operational and correctly assigned.

Figure 3: Trunk Configuration

```
Switch>show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,40

Switch>
```

Figure 4: Router Subinterface Configuration

```
Router0
Physical Config CLI Attributes

compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco IOS1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up

Router>show ip interface brief
Interface      IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0    unassigned      YES manual up            up
GigabitEthernet0/0.10 192.168.10.1    YES manual up            up
GigabitEthernet0/0.20 192.168.20.1    YES manual up            up
GigabitEthernet0/0.30 192.168.30.1    YES manual up            up
GigabitEthernet0/0.40 192.168.40.1    YES manual up            up
GigabitEthernet0/1    unassigned      YES unset  administratively down down
Vlan1            unassigned      YES unset  administratively down down
Router>
```

SECURITY IMPLEMENTATION (ACCESS CONTROL LISTS)

Security is one of the primary reasons for implementing VLAN segmentation in enterprise environments. To further strengthen the network, an Extended Access Control List (ACL) was configured on the router.

The ACL was designed to enforce departmental security policies and restrict unauthorized access to sensitive resources.

The following rules were implemented:

- Customer Service VLAN was denied access to the IT Department VLAN.
- Accounts Department VLAN was permitted access to the ATM/Server VLAN.
- IT Department VLAN was granted unrestricted access to all VLANs.
- General network communication was permitted where no restrictions existed.

ACL Policy Summary

Source VLAN	Destination VLAN	Action
Customer Service	IT Department	Deny
Accounts	ATM Server	Permit
IT Department	All VLANs	Permit
Others	Authorized Networks	Permit

Security Benefits:

- Prevents unauthorized access.
- Protects sensitive systems.
- Reduces insider threats.
- Enforces least-privilege access.
- Improves network security posture.

The ACL configuration was verified using:

show access-lists

Testing confirmed that Customer Service users could not access IT systems, while authorized departments could communicate successfully.

Figure 5: ACL Configuration and Verification

```
Physical Config CLI Attributes

Do not copy with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up

Router>configure terminal
      ^
% Invalid input detected at '^' marker.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
Router(config)#access-list 100 permit ip 192.168.30.0 0.0.0.255 any
Router(config)#access-list 100 permit ip any any
Router(config)#interface g0/0.10
Router(config-subif)#ip access-group 100 in
Router(config-subif)#
```

PROJECT VALIDATION AND TESTING RESULTS

After completing the network configuration, several tests were performed to verify functionality, connectivity, security, and compliance with project requirements.

VLAN Verification

Command Used:

show vlan brief

Result:

All VLANs were successfully created and assigned the correct switch ports.

Status: PASS

Trunk Verification

Command Used:

```
show interfaces trunk
```

Result:

The trunk link successfully carried VLAN 10, VLAN 20, VLAN 30, and VLAN 40 traffic.

Status: PASS

Router Verification

Command Used:

```
show ip interface brief
```

Result:

All router subinterfaces were operational and assigned the correct IP addresses.

Status: PASS

Connectivity Testing

Test 1:

Customer PC1 → Customer PC2

Result:

Successful communication.

Status: PASS

Test 2:

Customer PC1 → Accounts PC1

Result:

Successful communication through Inter-VLAN Routing.

Status: PASS

Test 3:

Accounts PC1 → ATM Server

Result:

Successful communication.

Status: PASS

Test 4:

Customer PC1 → IT PC1

Result:

Access denied by ACL.

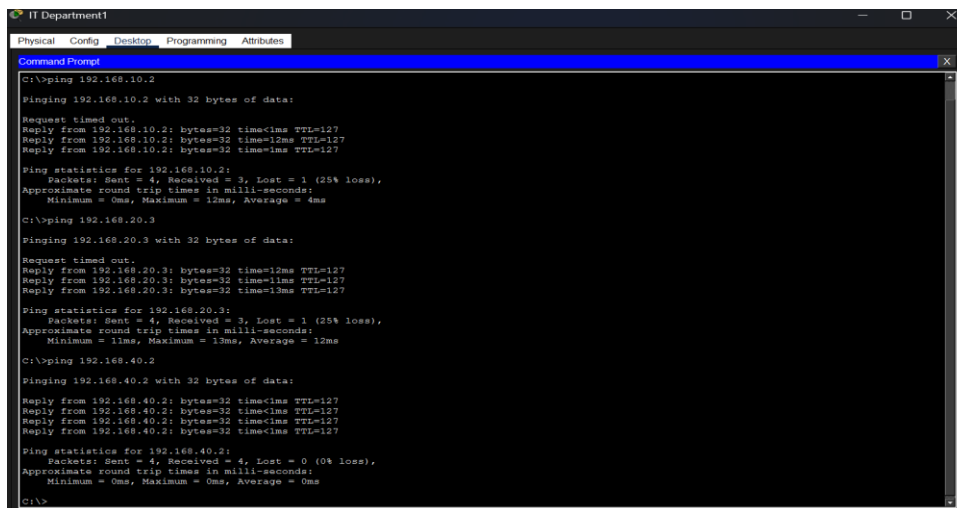
Status: PASS

Table 5: Validation Summary

Test	Result
VLAN Creation	PASS
Port Assignment	PASS
Trunk Configuration	PASS
IP Addressing	PASS
Inter-VLAN Routing	PASS
ACL Enforcement	PASS
Connectivity Testing	PASS
Security Policy Verification	PASS

The successful completion of all validation tests confirmed that the network was functioning correctly and securely.

Figure 6: Network Testing Results



```
IT Department1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.10.2
Pinging 192.168.10.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time=12ms TTL=127
Reply from 192.168.10.2: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 4ms
C:\>ping 192.168.20.3
Pinging 192.168.20.3 with 32 bytes of data:
Request timed out.
Reply from 192.168.20.3: bytes=32 time=12ms TTL=127
Reply from 192.168.20.3: bytes=32 time=11ms TTL=127
Reply from 192.168.20.3: bytes=32 time=13ms TTL=127
Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 13ms, Average = 12ms
C:\>ping 192.168.40.2
Pinging 192.168.40.2 with 32 bytes of data:
Reply from 192.168.40.2: bytes=32 time<1ms TTL=127
Reply from 192.168.40.2: bytes=32 time<1ms TTL=127
Reply from 192.168.40.2: bytes=32 time<1ms TTL=127
Reply from 192.168.40.2: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.40.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

CYBERSECURITY SIGNIFICANCE OF THE PROJECT

Network segmentation is a fundamental cybersecurity practice used by organizations to reduce risk and protect critical resources. This project demonstrates how VLANs and Access Control Lists can be combined to create a secure and manageable enterprise network.

Benefits of VLAN Segmentation

- Reduces broadcast traffic.
- Improves network performance.
- Limits lateral movement by attackers.
- Isolates sensitive departments.
- Enhances administrative control.

Benefits of Access Control Lists

- Restricts unauthorized access.
- Protects critical assets.
- Enforces security policies.
- Supports compliance requirements.
- Reduces insider threats.

Importance in Banking Environments

Financial institutions handle highly sensitive information including customer records, transaction data, ATM communications, and internal business operations. Unauthorized access to these resources can result in financial loss, reputational damage, and regulatory penalties.

The implementation of VLANs and ACLs provides multiple layers of security by ensuring that users only have access to the resources necessary for their job functions.

The project demonstrates several cybersecurity principles:

- Least Privilege
- Defense in Depth

- Network Segmentation
- Access Control
- Secure Resource Isolation

These principles are widely adopted in modern enterprise networks and cybersecurity frameworks.

CONCLUSION

This project successfully designed and implemented a secure enterprise network using VLANs, Inter-VLAN Routing, and Access Control Lists within Cisco Packet Tracer. The network was segmented into four logical departments: Customer Service, Accounts, Information Technology, and ATM/Server Infrastructure.

VLAN implementation provided effective network segmentation, reduced broadcast traffic, and improved overall security. Inter-VLAN Routing enabled controlled communication between departments while maintaining separation of network resources. Access Control Lists were applied to enforce security policies and restrict unauthorized access to sensitive systems.

Comprehensive testing and verification confirmed that VLAN creation, trunk configuration, IP addressing, Inter-VLAN Routing, and ACL enforcement were functioning as intended. Authorized departments successfully communicated with required resources, while restricted access policies were enforced correctly.

The project demonstrated the practical application of enterprise networking concepts and cybersecurity controls used in real-world organizations. Through VLAN segmentation, routing configuration, and access control implementation, the network achieved improved security, scalability, performance, and manageability.

Overall, the project successfully met all objectives and provided valuable hands-on experience in enterprise network design, implementation, troubleshooting, and cybersecurity best practices.